

2026

**College of Engineering
Student Advisory Board
Findings, Insights, & More!**

Agenda



- **Introduction/Mission Statement**
- **Department Insights & Recommendations:**
 - **BME**
 - **CHE**
 - **CME**
 - **CS**
 - **ECE**
 - **MIE**

Student Advisory Board



Maryam Abdul-Kader, Biomedical Engineering, *Senior*

Eli Burr, Computer Engineering, *Sophomore*

Luis Gonzalez, Electrical Engineering, *Sophomore*

Basil Hashim, Chemical Engineering, *Senior*

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Nathan Thokkudubiyapu, Computer Science, *Junior*

Taraz Tofighi, Industrial Engineering, *Senior*

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Mission Statement

To create awareness,
build a community, and
promote support and
opportunities to improve
the College of
Engineering student
experience.

Departmental Insights

Biomedical Engineering

Survey Overview



Objective

To characterize the career aspirations, professional backgrounds, program satisfaction, and self-assessed skill confidence of undergraduate students in the BME department.

Demographics

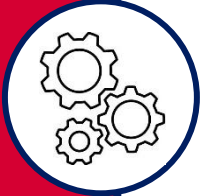
- 12% Response Rate
- 80% Upperclassmen
- Majority are involved in research & student orgs
- 22% do not have any extracurricular BME experience

Concentrations

- 36.6% Cell & Tissue
- 24.4% Imaging
- 19.5% Bioinformatics
- 19.5% Neural Engineering

Curriculum Leaves Students Underprepared for Industry

Students want deeper and continuous CAD/SolidWorks integration



- *"In Senior Design, I am the only one that knows SolidWorks... The only other time SolidWorks is taught is in BME 101, and most people forget it."*
- *"**Incorporate more SolidWorks/CAD** based courses. I have talked to countless students who wished there were SolidWorks/CAD courses beyond BME 101."*
- *"From getting an introduction to Arduino and SolidWorks in BME 101, **only to not use it again until senior year**. This semester, almost all of my classes revolve around circuits... yet we took one beginner class to circuits."*

The curriculum is too theoretical and lacks real-world application



- *"Mini group projects that **encourage research, prototyping, [and] application**... I feel is something the classes lack, as we generally only do this in BME 250 and Senior Design."*
- *"[The] **only class that actually matters** the most and I've learned greatly from... **is Senior Design**. I think the department should include this concept when students are in their junior year and so on."*
- *"I also wish the department would **state how research-focused our department is**... I just believe it would help people who are leaning towards industry make a better choice."*

Industry preparation is neglected compared to research or grad school



- *"[Faculty/advisors] as well as the classes just seem to be **pushing everyone towards research** or [a] masters in BME. This significantly dismisses those who want to go to industry post-grad... [T]here seems to be way **less help for those who need career guidance in industry**."*
- *"[There] are [a] **lack of opportunities** for those who hope to find internships/co-ops at a company unless you **find them on your own** fully outside of UIC."*
- *"A lot of the **classes are super vague** and cannot be applied to a lot of [fields]... to get more knowledge on it you have to either **do research or post-grad school**."*

Students Struggle With Variable Courses and Knowledge Gaps



Do the order and timing of core BME make sense and build effectively on each other?

54%

said “needs improvement”

Have you felt that your coursework connects theory to hands-on or project-based learning?

60%

said “occasionally”



Misalignment of Timing & Order

- “BME 399 Professional Development should be a 200-level course... **By junior/senior year, most students already understand** how to do interviews and prepare a resume.”
- “[Doing] circuits before understanding... was **easily overwhelming and relied too much on the TAs.**”
- “**I wish some classes were offered year-round** instead of only fall or only spring... **I cannot take BME 483 anymore** because it clashes with another class, and it is only offered in the spring.”



Repetitive Content & Inconsistency

- “I feel like a lot of the courses cover the same thing over and over again... [There are] **no set 'content'** that each class teaches, classes and what you learn... are **very instructor dependent.**”
- “Make sure the classes are taught the right way [because] the **professors change the curriculum** too much based on their strengths. Took 310 with [Professor] and it **did not help me at all** for 333.”
- “[There should be] a **more structured approach** when it comes to **core classes** and a more **clear guide on CAE courses.**”

Students Not Being Heard or Guided Effectively During Advising



Advising access and structural barriers



- "It is far **too hard to schedule** in-person meetings with advisors... my stated advisor... cancel[ed] meetings with me multiple times..."
- "[My advisor] was **unequipped to answer my questions**... It should not be this **difficult to connect** with the **needed administrators**."
- "**Clear labelling of prerequisites** and co-requisites everywhere and **not just the flowchart**."

70%

average undergrad satisfaction with their experience in BME at UIC

Advising clarity and trust



- "Better advising and **guidance based on your future goals** (e.g. graduating early, going into medical school, going into industry, other higher education, etc.)"
- "More **mandatory advising** throughout the year to **make sure** the student is **on track**."
- "Providing **more support** when it comes to **advising/planning** for the **next four years**."

only 20%

agreed student feedback is valued and acted upon in the BME department

Recommendations



Integrate Industry Preparation Throughout the Program

Embed career-relevant technical skills, early career exposure, and alumni/industry partnerships across all four years rather than concentrating them in senior year alone.

Strengthen Institutional Consistency

Establish shared curriculum standards across instructors and improve advising availability so students receive reliable, coherent support at every stage of their degree.

Invest More Into BME Community and Belonging

Students report strong attachment to the department as the highlight of their UIC experience; expanding mentorship programs, networking events, and peer connections will deepen that foundation further.



Chemical Engineering

Data Collection Overview



Objective

To understand student engagement and how **curriculum**, **resources**, and **community** influence the student experience.

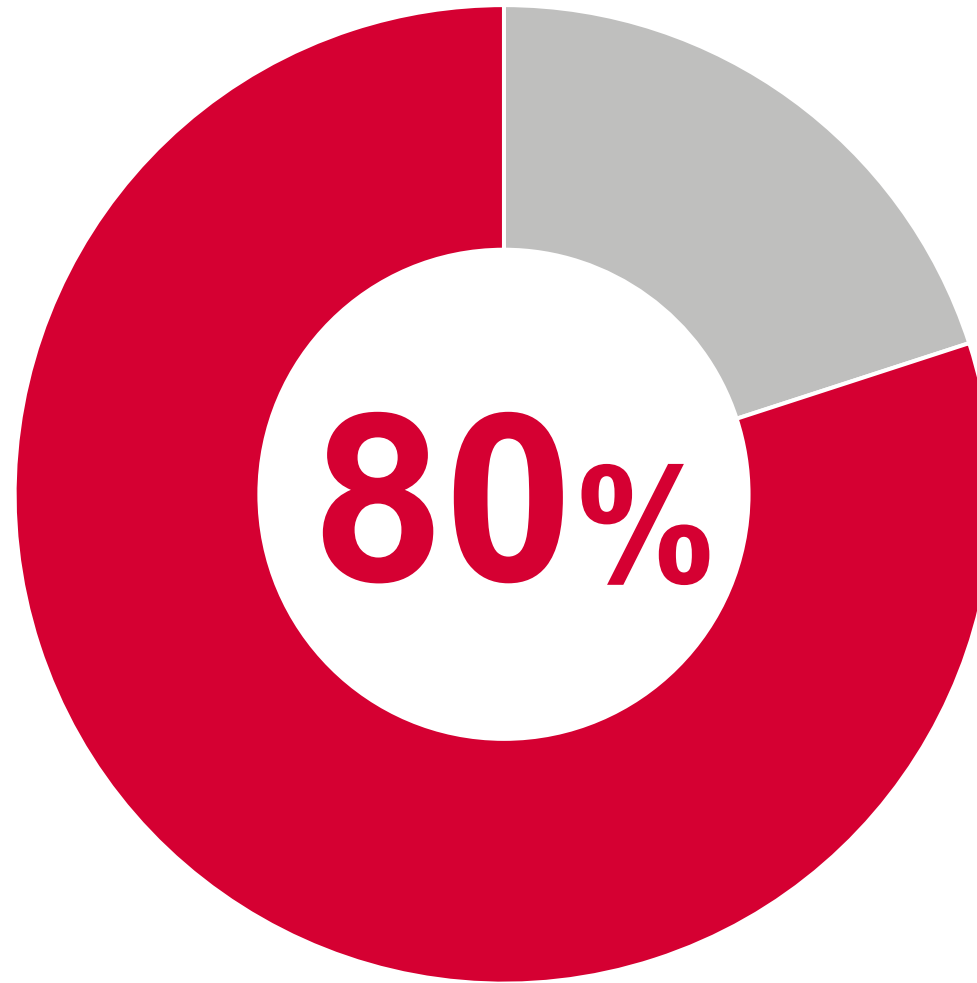
Methods

- Online survey
- Focus groups & Informal interviews

Participation

- ~35% of undergraduate population participated
- 15% transfer students
- Focus on juniors and seniors

Student Professional Goals



Industry is the dominant post-graduation path for students!

Curriculum Satisfaction Feedback



Students believe their classwork provides a strong fundamental basis in ChemE.

- “The curriculum is intensive for junior year students, prepare me with good work/ **study habit emphasis on consistency**, which might be valuable for industry”
- “Gives you real life expectations of how people **work in a team...**”
- “It’s **heavily theoretical** which makes sense that it **teaches us the fundamentals**. Would like to have more examples of real-world application like 313”



Students value transparency and flexibility within degree program.

- Students reported an average of **2.5/5 for degree planning ease**
- **Limited satisfaction with required class load** and pre/corequisite setup (constraining especially in senior year)
- Students suggested replacing certain required courses with more relevant alternatives
 - “I think there are a lot of unnecessary courses and bloat in our curriculum that could be replaced with things more beneficial to us, Achem Phem and CME 260 are all classes that are rather useless”



Students are eager to see a greater focus on **ethics, process safety, and practical troubleshooting** within the curriculum.

- Students reported an average of **3/5 for career-readiness**
- “...I believe it’s [the curriculum] **missing a course that emphasizes practical skills such as process operations, safety, troubleshooting, and industry tools**, which are essential ... in professional settings.”
- “wish we took more hands-on lab courses in the unit ops lab. We also don’t really learn about safety”
- “I feel the ChemE courses have gave me the knowledge and theory. But the issue is the application, **not many classes give us application besides CHE301 and CHE 433.**”
- “It’s heavily theoretical which makes sense that it teaches us the fundamentals. Would like to have more examples of real world application like 313”

Student Engagement with Career-Readiness Resources



ECC 50%

AIChE 38%

Percent of students who identified these as valuable resources!

Community and Informal Support Drive Student Success

All students surveyed reported **3/5 or higher** for sense of **belonging and comfortability** approaching departmental support staff.

Incoming students have repeatedly cited **sense of community as a reason for choosing UIC ChemE.**



Upperclassmen, professors, & staff are a large part of students' success networks.

Go-to place for career insight/help within CHE:

- "Upperclassmen", "Other Students mostly", "Peer advice and AChE events".
- "Professors", "Other Professors or alumni", "CHE Professors".
- "Front office, company presentations, peers", "Advisor/ECC", "Eduardo Martinez", "advisor/online UIC CHE page".



Study space may be a limiting factor for student engagement and community.

- "I think the students are fine, I feel as though some faculty members are unapproachable."
 - **Respondents clarified this is in-part due to limited office space**
- "The community is great. [It could improve with] ...**more space/lounge space for students.**"
- Junior students voiced concerns about overcrowding in existing lounge/library space in EIB

Recommendations



Create a more focused CHE study area with the capacity for TAs and professors to hold group sessions/office hours.

Students identified a need for stronger peer interaction and community-building opportunities outside the classroom.

Clarify/restructure curriculum and concentrations to create a more flexible, open-ended degree program.

Students reported confusion and limited flexibility in navigating elective options and planning their academic pathways.

Expand industry mentorship and project opportunities beyond senior design.

Students expressed interest in earlier exposure to real-world applications and industry engagement throughout the program.

We welcome continued discussion to refine and implement these recommendations!

Civil, Materials, & Environmental Engineering

Data Collection Overview



Objective

To understand the areas of opportunity in **resources** and **community engagement** for the CME student body.

Method

- Survey
 - Free response, Likert scale
- Informal Conversations

Participation

- **Small sample size**
- **11 participants**
- **Majority juniors, seniors, and commuters**

Students Desire Improved Resource Availability



Events are Held and Inaccessible Times

- *"Most students are available in the evening for extracurricular activities"*
- *"...[B]ecause UIC is a heavy commuter school the student body in general does not really take advantage of on campus resources as much as they can. A lot of the times students have classes during the morning and afternoon"*



Students are Unable to Access Tutoring and Office Hours

- *"There [are] many subjects for tutoring, but I think the hours allocated for this tutoring is subpar, [and] very lacking in the availability"*
- *"[T]utors, if available, should be open a little later than just during the day between classes"*

Students Desire Greater Focus on College Transition and Community Integration



Emphasis on Underclassman Involvement in Student Organizations



- *"[M]any clubs like concrete canoe and steel bridge are dedicated for upperclassmen because they've learned more topics. [Y]es anyone can go and learn but I wished there was more dedicated teaching"*
- *"Make the clubs less about projects and more about connecting and learning"*

Events Tailored to Underclassmen



- *"Have more events tailored to freshmen, sophomore, and transfer students who are still adjusting to the transition into the COE"*
- *"Have events tailored to grade levels especially freshmen that may be nervous to take advantage of these resources"*

Recommendations



Increase tutoring availability.

Commuter students gain increased access to academic help.

Create a mentorship program.

Connects underclassmen to upperclassmen, encourages community involvement, and provides an inclusive environment for underclassmen.

Computer Science

Objective

- Understand student engagement across **academic support, course structure, and experiential learning**
- Identify gaps in **accessibility, consistency, and clarity** within the CS student experience
- Inform **data-driven recommendations** for curriculum and support improvements

Methods

- **Online survey** distributed to CS undergraduate students
- **Qualitative responses** (open-ended feedback on courses, TAs, labs, etc.)
- **Mixed-method analysis:**
- Quantitative trends (confidence, workload, accessibility)
- Thematic coding (support gaps, TA consistency, lab experience)

Participation

- **79 undergraduate students** across all class years
- Majority representation from **sophomores, juniors, and seniors**
- Diverse perspectives:
 - Range of confidence levels (low → high)
 - Mix of commuters and on-campus students
- Strong representation from students in **core CS coursework**

Students consistently experience gaps in **academic support** and **course structure**, driving demand for more **reliable**, **structured**, and **accessible** learning systems.



Academic Support & Accessibility



- “Having more structured help sessions—**like debugging hours and exam reviews**—would make a huge difference in understanding key concepts.”
- “Office hours don’t always work for me as a commuter... having **more flexible or online options** would help a lot.”
- “Some **professors assume prior knowledge** and move too quickly, which makes it difficult for students encountering concepts for the first time.”

TA Experience & Consistency



- “**TA experiences vary a lot**—some are very helpful, while others are not helpful at all.”
- “There were times when **TAs didn’t show up** or were late, and we were left waiting without support.”
- “TAs should be **trained** more consistently... communication and reliability are just as important as technical knowledge.”

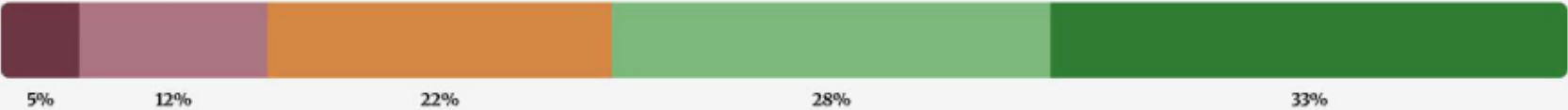
Labs & Learning [+ Experiential] Experience



- “**Labs feel like a ticking time bomb**—we’re rushing to submit rather than actually learning.”
- “Projects are helpful because we can work at our own pace, but **labs are stressful and time-pressured**.”
- “The best learning happens when concepts follow a clear flow—**lecture, then practice, then reinforcement through assignments**.”

Student Experience – Quantitative Insights

Academic support is accessible and helpful



Course expectations are clear and well-structured



Strongly Disagree Somewhat Disagree Neutral Somewhat Agree Strongly Agree

Even as most students report moderate satisfaction, nearly 1 in 3 continue to face challenges with support accessibility, workload, and course clarity.

Nearly 1 in 3 students experience gaps in support, clarity, and workload.

While up to 40–50% of students engage with available resources, a meaningful portion still experience gaps in academic support, course consistency, and access to industry-aligned, experiential learning opportunities.



STUDENTS ACTIVELY ENGAGING WITH ACADEMIC RESOURCES



45% Engaged

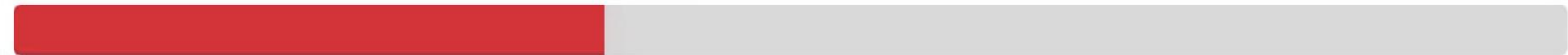
△ IDENTIFIED GAPS

Academic Support



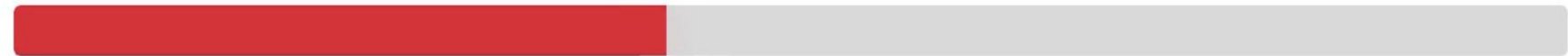
35% experiencing gaps

Course Consistency



38% experiencing gaps

Industry & Experiential Learning Access



42% experiencing gaps

Engagement exists — but consistency and access remain challenges.

Bridging academic structure with industry-aligned experiences is critical.

Recommendation



Standardize structured academic support systems

Implement weekly debugging sessions, exam reviews, and consistent TA support models to improve accessibility and reliability across courses.

Redesign course and lab experiences for consistency and clarity

Align teaching approaches, expectations, and lab structures to prioritize guided learning over time pressure and reduce variability across sections.

Expand industry-aligned and experiential learning opportunities

Integrate real-world tools, project-based learning, and earlier exposure to internships and technical preparation within the curriculum.

Electrical & Computer Engineering

Objective

- **Assess Support:** Evaluate faculty/TA accessibility and responsiveness.
- **Curriculum Review:** Measure grading consistency and theory-to-practice links.
- **Operations:** Identify pain points in scheduling and lab equipment.

Methods

- **Format:** 21-question anonymous digital survey.
- **Analysis:** Hybrid qualitative and quantitative data.
- **Timeline:** Collected November 2025 – January 2026.

Participation

- **Total:** 58 ECE student respondents.
- **Diversity:** Freshman through Graduate levels.
- **Scope:** From ECE 115 (Intro) to ECE 465+ (Advanced).

ECE students report that inconsistent communication and reactive course scheduling create significant barriers for planning.



Admin and Support

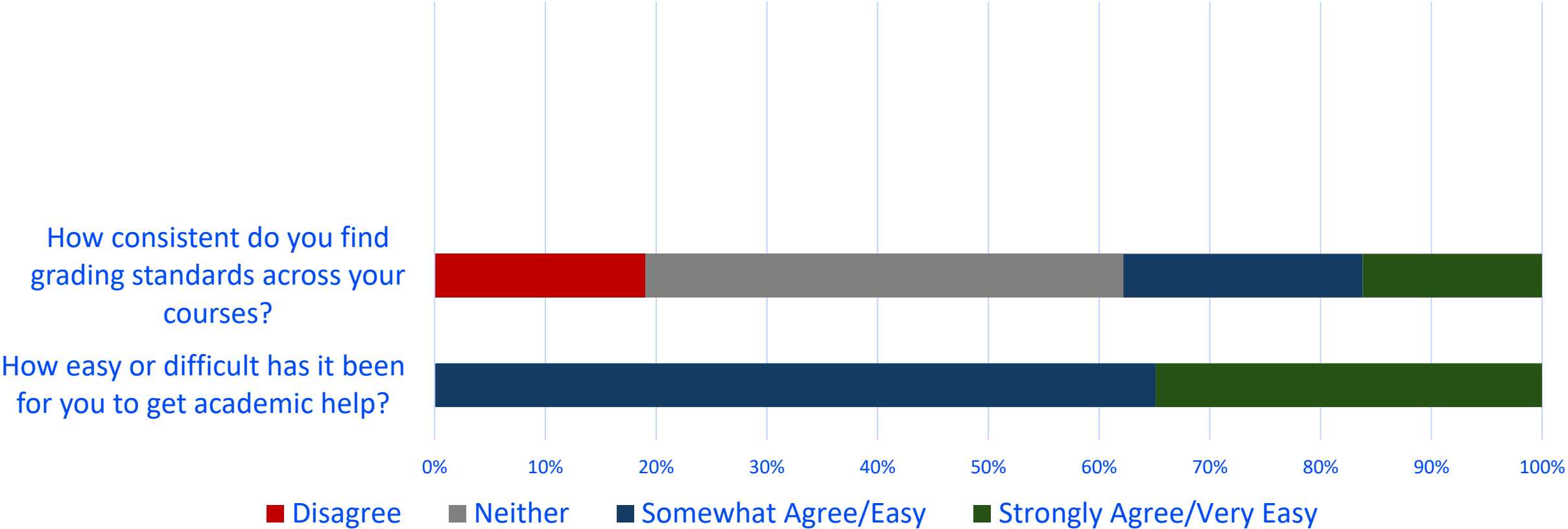
- *"Depending on the professor they have either been very responsive and helpful or not responsive and passive aggressive."*
- *"Advisors are unhelpful or do not know accurate information."*
- *"The quality of support is strongly dependent on the instructor, leading to inconsistent experiences across the department."*



Scheduling and Registration

- *"Create more sections EARLIER (weeks before the semester starts at least) so students don't have to scramble to register for the classes they need to take."*
- *"Open more seats for courses in ECE."*
- *"Scheduling is often a 'nightmare' because required sections are not opened early enough in the cycle."*

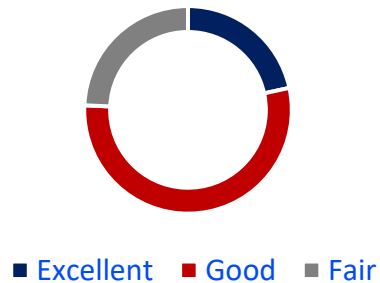
While overall satisfaction with the department remains positive, students report an inconsistent experience regarding grading standards.



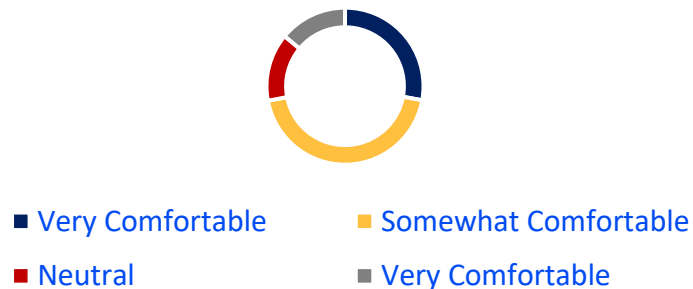
Students value hands-on learning but encounter equipment issues and a lack of synchronization with current lectures.



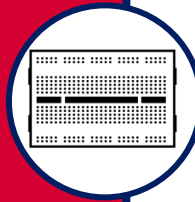
Class Material Connection



Approaching Faculty



Lab-to-Lecture Synchronization



- "Labs could be a little more in line with lecture material for some courses; sometimes we are doing things in lab we haven't touched in class yet"
- "The best-structured ECE courses are those where labs were balanced and pacing was good"
- "Make the labs easier to follow along with equipment that works"

Accessibility Barriers



- "Office Hours conflict with class schedule"
- "TA experiences vary a lot—some are very helpful, while others are not helpful at all"
- "I'm not sure of a solution but maybe a survey where everyone lists what courses they plan to take next semester could help"

Recommendation



Registration Planning

- Establish a mandatory pre-registration interest survey for all undergraduates two weeks before the registration cycle begins to proactively open and staff additional course sections based on predictive data rather than reacting to waitlists after the semester has already started.

Standardized Academic Quality

- Standardize the student experience by establishing department-wide grading rubrics for all core courses and mandating mid-semester synchronization audits to ensure lab assignments remain aligned with current lecture topics regardless of the specific instructor or TA assigned.

Hybrid Support Accessibility

- Transition to a hybrid-flexible academic support model by supplementing traditional mid-day office hours with twice-weekly virtual evening sessions via Discord or Zoom to accommodate the restrictive schedules of commuter students and those enrolled in foundational courses.

Mechanical & Industrial Engineering

Survey Demographics

Major:

- **87% Mechanical Engineering (63)**
- **13% Industrial Engineering (9)**
- **0% Engineering Management (0)**

Years in Attendance:

- **Less than 1: 50% (36)**
- **1: 13% (9)**
- **2: 19% (14)**
- **3: 13% (9)**
- **4 or more: 5% (4)**

COE Resources: Most students aren't using them, but those that do have overwhelmingly positive experiences



78%

Students "have never used"
most COE resources

59%

Have used and had a
positive experience with the
Makerspace

Key Quotes

- "I used the **Makerspace** for a school assignment and it **came out perfect** and they held the item for almost a week because I kept forgetting to go grab it so that was very nice. Also when I went to the **Counseling Center**, I was able to stay **on track for graduation** during the advising appointment."
- "The people in the **Makerspace** were very helpful, and it has **taught me skills** that I think would be useful in the future."
- "UIC, particularly the **Makerspace** and my student org have taught me so much about the **manufacturing industry** and allowed be to **obtain skills** which have **secured me jobs and internships**"

Career Readiness: Students want to know more about the field early on and need more support from UIC



26%

"have a basic sense of the kinds of careers" MIE students pursue

38%

agree that UIC has helped them "find opportunities within or outside of the University that directly correlated to my career interests"

Key Quotes

- *"GPIP was helpful for gaining experience. The onboarding process however, was really terrible. Finally, I wish GPIP was offered to transfer students in the first semester."*
- *"I've used the Engineering Career Center to get advice on how to improve my resume. I've also utilized the ELC almost weekly for help with Financial Engineering."*
- *"I just don't know what fields Mechanical Engineering pursue other than defense companies."*

AI: More beneficial for professors to model responsible use rather than discouraging or outright banning it



96%

"occasionally or regularly"
use AI to assist them with
homework and/or exam
preparation

8%

"occasionally or
regularly" use AI to
do their homework
for them



Key Quotes

- "People using AI at this point is **inevitable** so I would suggest that there just be **requirements or restriction** on assignments..."
- "AI is **useful as a tool** to better understand class notes, to **fill in missing gaps** in comprehension. **Faculty should model** to students **how to responsibly leverage AI** during study instead of outright prohibiting it, my belief is that **strict prohibitions lead to AI abuse** on assignments and exams."
- "I think the professors **shouldn't completely discourage** the use of AI but there should always be the warning of "do not copy from it" because, especially in engineering, AI models do a really poor job of giving accurate information; AI should be **used more as a guide** to find what to actually research rather than to use it as an answer key and it would be nice if professors kind of looked at how to do that and presented it to students since there is **no way to actually control** the use of it."
- "**Hands-on experience and simulations** in class or hw that **AI isn't capable of** is helpful for the student and knowledge of what it is to be an engineer. AI is going to be **unavoidable for academics** so implement it with academic and make it be useful."
- "The usage of AI can actually **really help the learning process**. I have used it to elaborate on concepts that I don't understand or haven't had the time to attend office hours for. If students **use it as a study tool** it can be very helpful."

Recommendations



- Existing COE resources such as the Makerspace are benefitting students. Continue to bolster them and better promote their use, formally and informally.
- Students would like to feel connected to their careers from Day 1. Encourage professors to discuss how their particular courses connect to the industry.
- Virtually every student is using AI in some capacity to assist them with studying, exam preparation, or conceptual reinforcement. Professors should model how AI can be used to support success in their class, rather than outright penalizing its use.

**Thank You For Your Time
and Commitment to UIC!**

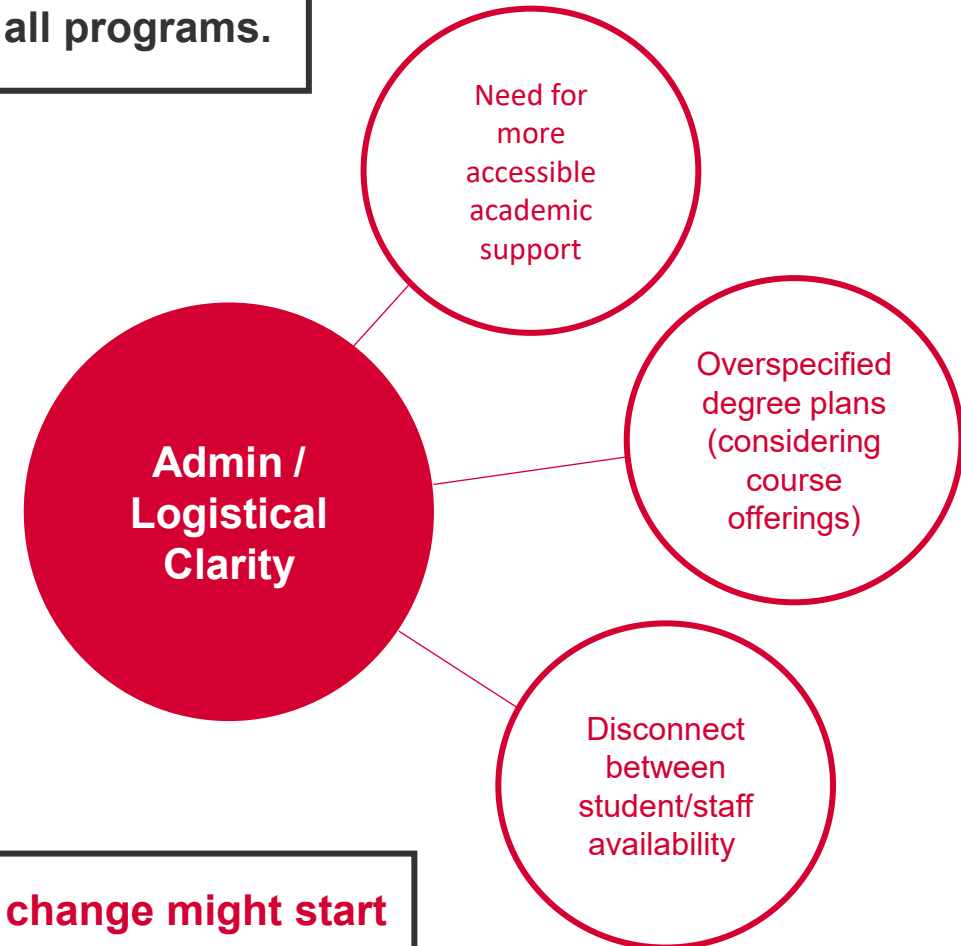
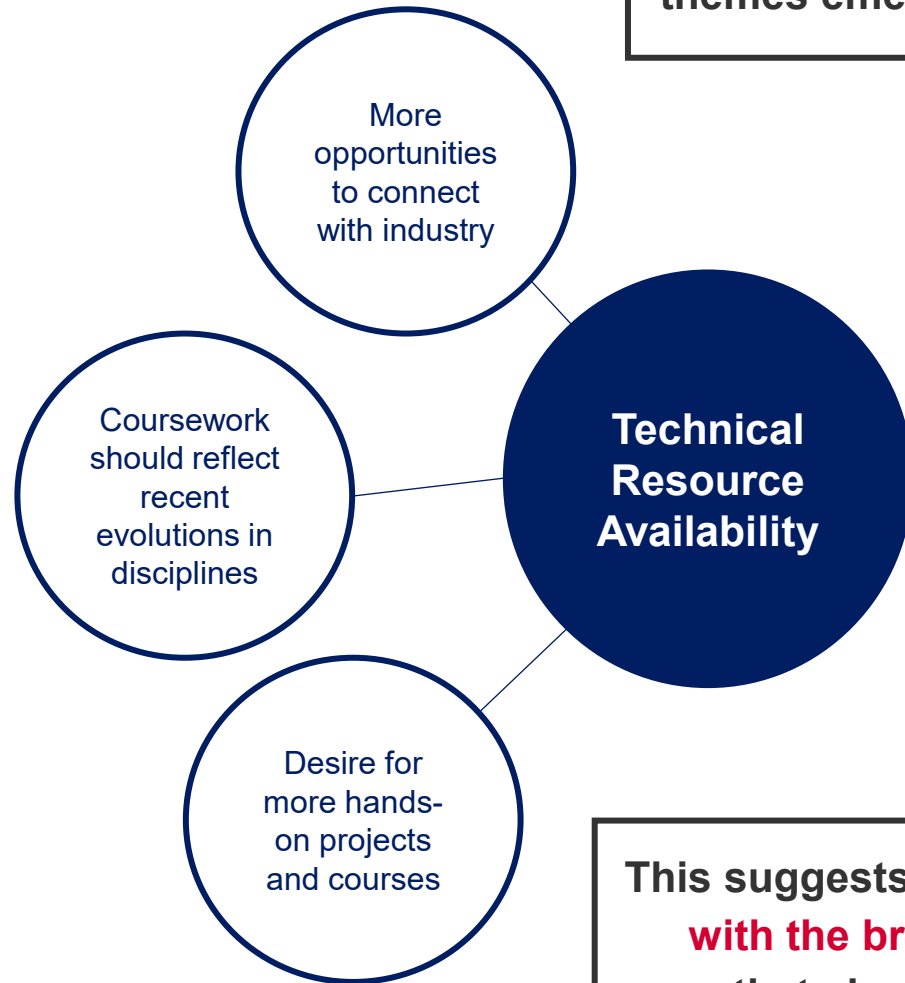
Supplemental Material

College-Level Insights

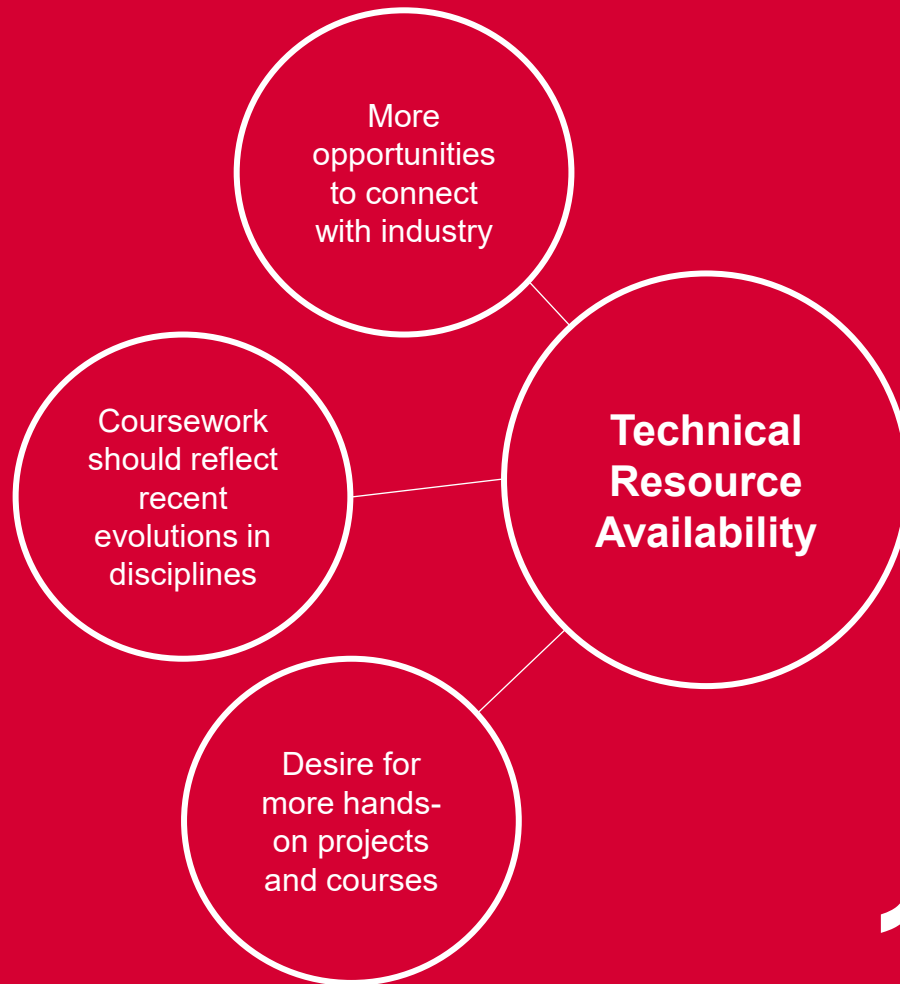
Overview



While each department highlighted unique challenges, several consistent themes emerged across all programs.



This suggests **meaningful change might start with the broader, college-wide systems** that shape the student experience!



1. Technical Resources

Recap: Current Resource Usage



Observed Underutilization of Resources

- A significant portion of the student body does not regularly use formal departmental resources, with a large number of students not attending faculty office hours.
- Students have either heard of UIC/COE resources or haven't, but in either case haven't used them.
 - Potentially more of an issue of access as opposed to an information issue

Trends in Current Student Engagement

- Students generally reported wanting more *structured* career opportunities
- However, they also reported heavy reliance on informal support
 - This includes: peer advice, upperclassmen mentorship, and student organizations

Areas of Continued Interest



Lab Courses

- In several departments students believe the value of lab courses are limited by the format of the class
 - Inconsistency between sections and focus on “outdated” methods were the most common comments

Departmental Professional Development

- Students believe that required professional development courses are valuable
 - Earlier implementation may bolster its relevance to students discovering their career paths

Student Organizations

- Students have been shown to engage with student organizations, but still want more opportunities to participate in them
 - Call for more department or specific-industry focused clubs
- Lack of clarity in communications with student orgs
 - Purchasing guidelines are not proactively communicated
- Not well integrated into college, making recruiting difficult

Recommendations



Incentivize participation in student orgs

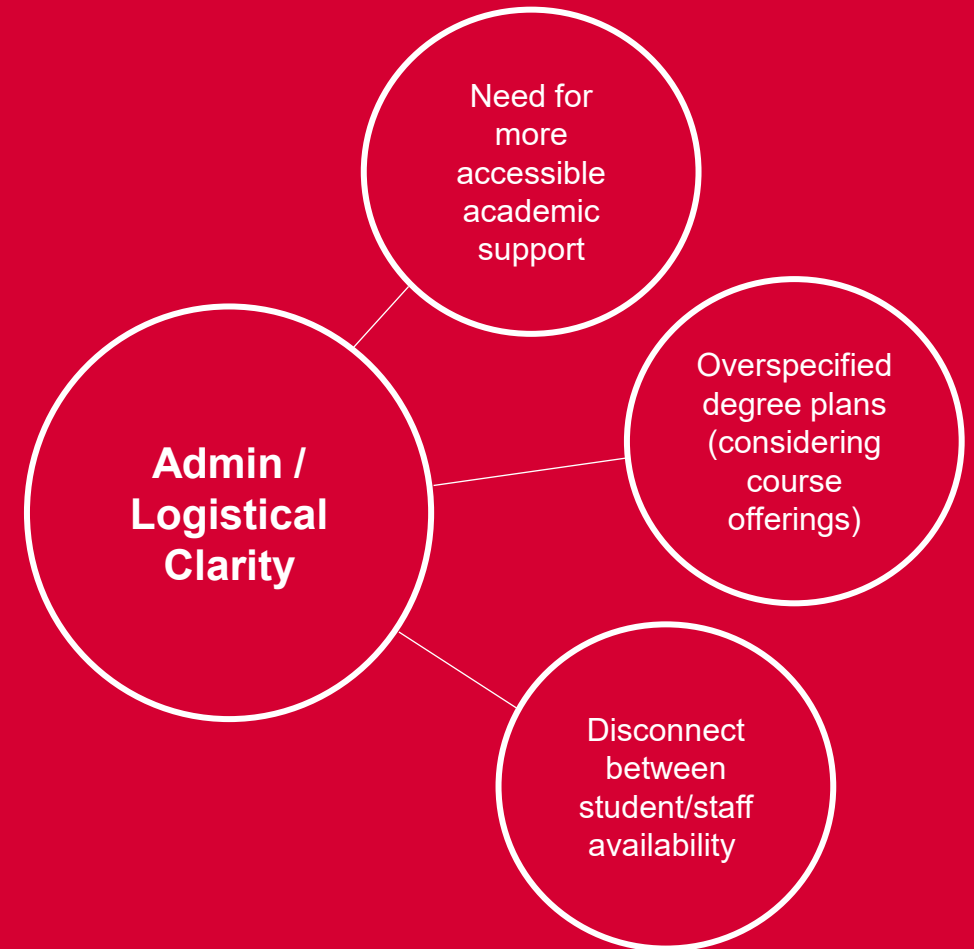
- Increasing engagement with student orgs has the potential to address student desire for more industry-relevant technical experiences
- Students want to see more student orgs for specific things, clearer documentation of and communication from administration about running an org may bolster club participation/creation

Continue to engage students through ECC

- Students currently view the ECC as an important hub for information, publishing more specific career resources through it may be beneficial

More industry exposure for students

- More focus on co-ops and other experiential learning experiences
- Greater collaboration between UIC and other regional resources (ex. mHUB)



2. Administration/Curriculum

Overall Comments on Curriculum

- **Disconnect between senior design experience and prior/prerequisite courses**
- **Several bottlenecks within degree programs not communicated effectively**
- **Limited class availability**

Lack of Clarity in Advising & Course Expectations

- **Not clear who to approach with specific issues**
 - **Advisors are unhelpful or do not know accurate information**
 - **Distinction between course/professional advising not made clear, preventing students from seeking advising**
- **Unclear learning expectations of students, particularly concerning AI**

Recommendations



Improve the structure of coursework advising

Invest more resources in the coursework advising process, particularly for transfer students at the very beginning of their time at UIC.

Expand core-engineering course options

Allow more options for college-wide course requirements so students can gain more specialized knowledge, expand interdisciplinary knowledge exchange, and minimize scheduling conflicts for commuter student population.

Key Takeaways



- **There are consistent gaps between the skills students are learning and the skills they require for industry success.**
- **Time constraints and a lack of clarity prevent students from engaging with resources.**
- **Curriculum is restrictive and courses are often lacking in availability.**
- **Deficient communication between students and administration leads to uncertainty within the student body.**

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